

SimCSE Reproduction: Final Research Report

Implementation-level reproduction, reduced-data scaling, controlled ablations, alignment/uniformity analysis, and discrepancy analysis.

Abstract

This project presents an implementation-level reproduction and resource-constrained experimental study of unsupervised SimCSE using BERT-base-uncased. The completed study validates the training pipeline, evaluates raw BERT and the official pretrained reference, and examines 10K/25K/50K training, dropout, temperature, and representation geometry. The strongest local result is 76.38% STS-B with independent dropout 0.20. The study is a validated reduced-data reproduction and mechanism study, not a claim of completing the original 1M-sentence paper-scale training.

Main Results

Experiment	Size	STS-B
Raw BERT CLS	—	30.18%
SimCSE	10K	53.55%
SimCSE	25K	62.59%
SimCSE	50K	69.07%
No dropout	50K	60.21%
Fixed mask 0.10	50K	41.07%
Independent dropout 0.20	50K	76.38%
Temperature 0.01	50K	68.88%
Temperature 1.00	50K	58.33%
Official pretrained, local STS-B	—	82.12%

1. Research Questions

The study asks whether unsupervised SimCSE improves over raw BERT, how performance changes with reduced training data, whether independent stochastic dropout matters, how temperature affects performance, and how alignment/uniformity relate to downstream STS-B.

2. Scaling

STS-B rises from 53.55% at 10K to 62.59% at 25K and 69.07% at 50K. The incremental gains are +9.04 and +6.48 percentage points. This is an empirical diminishing-return pattern over the tested range, not a universal scaling law.

3. Dropout

No dropout reaches 60.21%, a fixed shared mask reaches 41.07%, and independent dropout at 0.20 reaches 76.38%. The fixed-mask result supports interpreting independent stochastic perturbation as an important part of the positive-view mechanism.

4. Temperature

Temperature 0.01 reaches 68.88% and temperature 1.00 reaches 58.33%, compared with the 0.05 baseline. The completed evidence demonstrates material temperature sensitivity but does not establish a universal optimum.

5. Geometry

No-dropout has the strongest local uniformity but only 60.21% STS-B. Temperature 1.00 has the strongest alignment but only 58.33% STS-B. These observations show that neither metric alone is sufficient; the results support an alignment–uniformity balance interpretation.

6. Reference correction

The published 76.25% figure is Avg. STS across seven datasets, not STS-B. Therefore the local 76.38% STS-B result must not be described as beating 76.25%. The local 82.12% official-checkpoint result is also an STS-B validation result and is not directly equivalent to the published seven-task average.

7. Limitations

The study does not complete 1M-sentence training or the full seven-task benchmark. The local reduced-data checkpoint selection does not exactly reproduce official best-checkpoint selection. A complete pooling ablation is also not established.

8. Conclusion

We reproduced and validated the core unsupervised SimCSE training implementation and characterized its learning behavior, mechanism-level ablations, and representation geometry under a resource-constrained reduced-data training regime.

Geometry Table

Experiment	STS-B	Alignment	Uniformity
Raw BERT	30.18%	0.1771	-1.0560
SimCSE 50K	69.07%	0.2924	-2.5705
No dropout	60.21%	0.4486	-2.7144
Fixed mask 0.10	41.07%	0.4057	-2.3718
Dropout 0.20	76.38%	0.1946	-2.3814
Temperature 0.01	68.88%	0.1642	-1.5738
Temperature 1.00	58.33%	0.1030	-1.9329

Final scientific position: the completed evidence supports a validated implementation-level reproduction plus a controlled mechanism study. It does not support a claim of full paper-scale reproduction.

See the accompanying Claim Audit for a statement-by-statement distinction between measured results, source-supported facts, inference, limitations, and claims not established.